Power BI Project Themes and Ideas

Theme 1 – Stock Market Analytics

- **Domain:** Finance & Investment
- **Storyline:** Students build a complete equity-analysis workflow — from cleaning historical stock data to designing professional investor dashboards.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Cleaning and Preparing Stock Data	Clean daily stock prices (open, high, low, close, volume). Remove missing dates and calculate daily returns.	Daily Return %, Moving Average (7-day, 30-day)
Mini Project 2	Visualizing Market Trends	Compare multiple companies (TCS, Infosys, Reliance) using time-series and sector charts.	Avg Return, Volatility, Top Performing Sector
Capstone Project	Investment Portfolio Analytics	Merge index and company data; apply DAX for ROI, Sharpe Ratio, and Risk Score; create investor dashboard.	Portfolio Return, Sharpe Ratio, Beta, Risk-Adjusted Performance

- **Skills Covered:** Power Query Cleaning, Time-Series Modeling, DAX KPIs, Financial Dashboard Design
- **Dataset Source:** Yahoo Finance (yfinance API) | Kaggle Stock Market Data

Theme 2 – FinTech Product Analytics

- **Domain:** Financial Technology / Digital Payments
- **Storyline:** Analyze transaction data from a FinTech app to understand user behavior and business performance.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Transaction Data Cleanup	Identify failed and duplicate transactions; fix timestamps and standardize payment modes.	Transaction Success Rate, Avg Transaction Value
Mini Project 2	User Engagement Dashboard	Visualize daily users, frequency of transactions, and retention patterns.	Active Users %, Retention Rate, Avg Monthly Spend
Capstone Project	Business Growth & Revenue Insights	Integrate transaction + user feedback data; calculate MRR, Churn Rate, CLV; build multi-page dashboard.	MRR, Churn %, CLV, Revenue Growth %

Skills Covered: Data Integration, Customer Segmentation, DAX for MRR & CLV, Business Storytelling

Dataset Source: Kaggle PaySim Transactions

Theme 3 – E-Commerce Sales Insights

Domain: Retail & Consumer Analytics
Storyline: Develop insights for an online store by analyzing sales, returns, and profitability across regions and categories.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Cleaning Sales Data	Handle missing invoice dates, duplicate entries, and product category mismatches.	Total Orders, Avg Order Value
Mini Project 2	Regional Performance Dashboard	Visualize sales by region and category with interactive filters and trendlines.	Revenue Growth %, Top Selling Category, Return Rate
Capstone Project	Profitability & Customer Insights	Merge customer demographics with sales data for RFM analysis and retention strategy.	Customer Lifetime Value, Profit Margin %, Repeat Rate

Skills Covered: Data Modeling, RFM Analysis, Power BI Visuals, Storytelling for Retail
Dataset Source: Kaggle E-Commerce Dataset

Theme 4 – Hospital Patient Analytics

Domain: Healthcare & Operations
Storyline: Improve hospital efficiency by analyzing patient records, doctor performance, and treatment patterns.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Patient Record Preparation	Clean data with missing ages/genders and merge doctor assignments.	Avg Age, Missing Rate %, Patient Volume
Mini Project 2	Treatment & Doctor Dashboard	Visualize admissions, doctor load, and disease distribution across departments.	Avg Stay Length, Patients per Doctor, Top Diseases
Capstone Project	Hospital Efficiency Optimization	Combine treatment + feedback data; create DAX KPIs for Satisfaction Index and Cost per Patient.	Efficiency Score, Patient Satisfaction %, Readmission Rate

Skills Covered: Data Cleaning, Healthcare KPIs, DAX Index Creation, Operational Dashboard Design
Dataset Source: Kaggle Hospital Patients Data

Theme 5 – Air Quality & Pollution Dashboard

Domain: Environmental Analytics
Storyline: Study pollution levels across cities and time to inform sustainability and policy decisions.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Cleaning Air Quality Readings	Handle missing sensor values and standardize city names and dates.	Avg PM2.5, Missing Data %, City Count
Mini Project 2	Regional Pollution Trends	Create maps and trendlines to compare AQI across cities and seasons.	AQI Trend, Top 5 Polluted Cities, Seasonal Change %
Capstone Project	Sustainability Insights Dashboard	Integrate AQI + weather data and visualize correlation with population density.	Avg AQI per State, Clean-Air Improvement %, Policy Impact Index

Skills Covered: Data Merging, Geospatial Visualization, Map Layers, Sustainability Analytics
Dataset Source: OpenAQ API | Kaggle Air Quality Data India

Theme 6 – COVID-19 Global Tracker

Domain: Health & Epidemiological Analytics
Storyline: Build a real-time dashboard to track infection trends, vaccination rates, and regional comparisons worldwide.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Global COVID Data Cleaning	Clean and prepare country-level daily cases, deaths, and vaccination counts.	Active Cases %, Daily New Cases, Mortality Rate
Mini Project 2	Pandemic Visualization Dashboard	Create continent-wise trendlines, maps, and KPIs for infection vs recovery.	Infection Rate %, Vaccination Progress %, Recovered Cases
Capstone Project	Pandemic Impact Analytics	Merge economic indicators (GDP, unemployment) with health data to analyze pandemic effects.	Recovery Ratio, GDP vs Cases Correlation, Vaccination Efficiency

Skills Covered: Data Integration, Geo-Maps, Comparative Analytics, Power BI Service Auto-Refresh
Dataset Source: Our World in Data COVID Dataset

Theme 7 – Education Performance Dashboard

Domain: Education & Learning Analytics
Storyline: Analyze academic performance, attendance, and subject-wise improvement to assist school administrators.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Student Marks Cleanup	Handle missing marks, standardize subjects, and merge attendance sheets.	Avg Marks, Attendance %, Pass Rate
Mini Project 2	Academic Trends Dashboard	Create visuals for class-wise and subject-wise performance.	Top Subjects, Gender-wise Avg Marks, Overall Grade Trend
Capstone Project	Learning Insights Dashboard	Combine multiple years' data to track improvements and teacher impact.	Performance Growth %, Teacher Impact Index, Improvement Ratio

Skills Covered: Data Cleaning, Relational Modeling, Education KPIs, Conditional Formatting
Dataset Source: Kaggle Students Performance Dataset

Theme 8 – Bank Loan Analysis

Domain: Finance & Banking Analytics
Storyline: Evaluate loan portfolio performance, credit risk, and default trends using customer-level financial data.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Loan Application Data Preparation	Clean records with null credit scores and inconsistent dates.	Approval %, Avg Loan Value, Missing Score %
Mini Project 2	Loan Performance Dashboard	Visualize approved vs rejected loans, region-wise patterns, and NPA %.	Default Rate, Avg Credit Score, High-Risk Region
Capstone Project	Predictive Credit Risk Insights	Combine customer + loan + repayment data; apply DAX to find profitability and risk segments.	Risk Index, Profit by Loan Type, Recovery %, NPA Trend

Skills Covered: Power Query Joins, DAX Ratios, Risk Modeling, Banking Dashboards
Dataset Source: Kaggle Bank Loan Modelling Dataset

Theme 9 – Startup Funding Insights

Domain: Entrepreneurship & Investment Analytics
Storyline: Explore India's startup ecosystem, funding rounds, and investor preferences to uncover trends.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Startup Funding Data Preparation	Clean raw startup data with inconsistent funding types and missing city info.	Total Funding, No. of Startups, City Count
Mini Project 2	Investor Trend Dashboard	Visualize top investors, popular sectors, and funding stages over time.	Top Sectors, Avg Deal Size, Funding Growth %
Capstone Project	Investment Insights Dashboard	Integrate multi-year data; calculate CAGR, city growth, and funding success probability.	CAGR %, Investor Participation Rate, Funding-to-Exit Ratio

Skills Covered: Power BI Time Intelligence, CAGR Calculations, Investment Analytics, Data Storytelling
Dataset Source: Kaggle Indian Startup Funding Data

Theme 10 – Crypto Market Analytics

Domain: Blockchain & Digital Assets
Storyline: Examine cryptocurrency trends, volatility, and portfolio risk across top coins.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Crypto Price Data Cleaning	Import live data via API, handle missing timestamps, calculate % change.	Daily Change %, Avg Volume, Volatility
Mini Project 2	Crypto Trend Dashboard	Visualize price, volume, and market dominance for top 10 coins.	Market Cap %, Top Performing Coin, Avg 30-day Return
Capstone Project	Portfolio Performance Insights	Combine multi-coin data to build an investor portfolio and risk dashboard.	Portfolio ROI, Sharpe Ratio, Risk Score, Avg Holding Return

Skills Covered: API Integration, Real-time Data Modeling, DAX Time Measures, Financial Dashboards
Dataset Source: CoinGecko API | CoinMarketCap Dataset

Theme 12 – Retail Store Performance Dashboard

Domain: Retail & Sales Analytics
Storyline: Analyze multi-store sales performance to uncover profitability, seasonality, and top products.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Sales Data Cleaning	Standardize product categories, fix date formats, and merge sales with returns data.	Total Revenue, Avg Discount %, Missing Sales %
Mini Project 2	Store-wise Sales Dashboard	Create visuals for top regions, categories, and seasonal sales peaks.	Profit Margin %, Top Store, Revenue Growth %
Capstone Project	Regional Profitability Insights	Add demographics and target analysis; identify top 10 profitable zones.	Avg Order Value, Sales per Category, YoY Growth %

Skills Covered: Power Query Merging, Date Modeling, DAX Profit Metrics, Report Design
Dataset Source: Kaggle: Superstore Dataset

Theme 13 – HR Attrition Dashboard

Domain: Human Resource Analytics
Storyline: Help HR teams predict attrition and understand employee engagement through data.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Employee Data Preparation	Clean employee records, merge departments, and calculate tenure.	Avg Tenure, Missing Values %, Gender Ratio
Mini Project 2	Attrition Analysis Dashboard	Visualize employee attrition by age, department, and salary.	Attrition %, Avg Monthly Income, Dept-wise Turnover
Capstone Project	Predictive Retention Insights	Combine survey and performance data to build a retention prediction model.	Retention %, Job Satisfaction Index, Promotion Gap %

Skills Covered: HR Metrics, Attrition Modeling, DAX Ratios, Power BI Conditional Formatting
Dataset Source: Kaggle: IBM HR Attrition Dataset

Theme 14 – Movie Revenue Insights

Domain: Entertainment & Media Finance
Storyline: Study how budget, rating, and genre affect box-office success.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Movie Data Cleaning	Fix null budgets, handle missing revenue values, standardize release years.	Missing % per Field, Avg Budget, Record Count
Mini Project 2	Movie Trends Dashboard	Show revenue by genre, rating, and production company.	Avg ROI, Top 5 Genres, Profit Margin %
Capstone Project	Box Office Success Analytics	Merge IMDb or TMDB ratings and predict ROI based on features.	ROI %, Rating Correlation, Profit per Budget Unit

Skills Covered: Profitability Analysis, Data Joins, Visual Correlation Charts, Financial Dashboards
Dataset Source: Kaggle: TMDB Movie Data

Theme 15 – Energy Consumption Tracker

Domain: Energy & Sustainability Analytics
Storyline: Track energy usage and renewable source contributions to promote sustainable development.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Power Usage Data Cleaning	Clean energy readings and convert timestamps; fix units (kWh).	Avg Consumption, Peak Usage, Missing Data %
Mini Project 2	Renewable vs Non-Renewable Dashboard	Visualize renewable energy shared by city/region.	Renewable %, Avg Daily Output, Top Energy Source
Capstone Project	Sustainable Energy Analytics	Combine weather + consumption data for energy optimization insights.	Energy Efficiency %, Renewable Trend %, CO2 Saved (tons)

Skills Covered: Time-Series Analytics, Geo Visuals, KPI Tracking, Energy Forecasting
Dataset Source: Kaggle: Electric Power Consumption Dataset

Theme 16 – Airline Performance Dashboard

Domain: Aviation & Transport Analytics
Storyline: Evaluate flight punctuality, route performance and customer experience to optimize operations.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Flight Data Cleaning	Fix invalid departure/arrival times and remove cancelled duplicates.	Total Flights, Cancelled %, Avg Delay (min)
Mini Project 2	Route Performance Dashboard	Visualize top routes, on-time percentages and delay trends.	On-time Rate %, Delay Causes Breakdown, Load Factor
Capstone Project	Operational Efficiency Analytics	Integrate customer feedback + route profit data to find profit vs delay trade-offs.	Avg Profit per Route, Customer Satisfaction %, Delay Impact Index

Skills Covered: Data Cleaning, Time Analysis, Geo-Maps, Correlation Visualization
Dataset Source: Kaggle Flight Delays Dataset

Theme 17 – Tourism & Hotel Analytics

Domain: Hospitality & Tourism Management
Storyline: Study hotel bookings, seasonality and cancellations to improve occupancy and revenue.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Booking Data Preparation	Clean guest bookings and handle duplicate records and missing dates.	Total Bookings, Cancelled %, Avg Stay Nights
Mini Project 2	Hotel Revenue Dashboard	Visualize monthly revenue, ADR and occupancy rate by market segment.	Occupancy %, ADR, RevPAR, Cancellation Ratio
Capstone Project	Tourism Forecast Dashboard	Merge weather + booking data to forecast demand and revenue.	Forecasted Occupancy %, Demand Index, Revenue Growth

Skills Covered: Forecasting with DAX, Time Series Visuals, Hospitality Metrics (ADR, RevPAR)
Dataset Source: Kaggle Hotel Booking Demand Dataset

Theme 18 – Sports Performance Dashboard

Domain: Sports Analytics
Storyline: Compare team and player performances across seasons to identify winning patterns.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Player Stats Cleaning	Fix null scores, standardize player names, merge season tables.	Total Matches, Avg Score, Strike Rate
Mini Project 2	Team Performance Dashboard	Visualize team wins, losses and player impact metrics.	Win %, Player Contribution %, Top Scorers
Capstone Project	Season Analytics & Predictive Insights	Integrate multi-season data to predict team success factors.	Win Probability, Consistency Index, Performance Trend

Skills Covered: Data Modeling, Sports KPIs, Correlation Charts, Predictive Visuals
Dataset Source: Kaggle IPL Dataset | Kaggle FIFA Player Data

Theme 19 – Crime Rate Analysis

Domain: Public Safety & Governance
Storyline: Visualize crime distribution by state and type to assist policy decision-making.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Crime Data Preparation	Clean state-wise crime records and standardize categories.	Crime Rate %, Population per Crime, Data Completeness %
Mini Project 2	Crime Visualization Dashboard	Build maps and bar charts for offence type distribution.	Top Crime Type, Year-on-Year Change %, Hotspot Regions
Capstone Project	Crime Trend & Policy Analytics	Integrate socio-economic factors (GDP, Education) to correlate crime rates.	Crime Index, Socio-Crime Correlation %, Crime Drop Rate

Skills Covered: Geo Visualization, Data Blending, Comparative Analytics, Policy Metrics
Dataset Source: Kaggle Crime in India Dataset

Theme 20 – Real Estate Market Dashboard

Domain: Property & Investment Analytics
Storyline: Analyze property prices and market trends to support real estate investment decisions.

Stage	Project Title	Description	Sample KPIs
Mini Project 1	Property Listing Data Cleanup	Fix missing prices and standardize locations and property types.	Avg Price/sq ft, Property Count, Missing Data %
Mini Project 2	Market Price Trend Dashboard	Visualize property price trends by city and type.	City-wise Avg Price, Yearly Growth %, Top Expensive Areas
Capstone Project	Investment Opportunity Analytics	Integrate rental + sales data to compute yield and ROI.	Rental Yield %, ROI %, Price Growth Rate

Skills Covered: Spatial Analytics, Real Estate KPIs, ROI Calculation, Interactive Maps
Dataset Source: Kaggle House Sales Prediction Dataset